Project Design Phase
Proposed Solution Template

Date	2 March 2026
Team ID	MAB-041
Project Name	Heart Disease Analysis using Tableau
Maximum Marks	2 Marks

Proposed Solution Template:

The proposed solution is an end-to-end data analytics system that leverages Tableau for visualizing heart disease data and generating actionable insights. The system is designed to transform raw healthcare data into interactive dashboards that highlight key trends, correlations, and risk factors. The solution begins with collecting the heart disease dataset and storing it in a relational database such as MySQL. SQL queries are then used to clean, filter, and structure the data to ensure consistency and readiness for analysis. This preprocessing step ensures that the visualizations are accurate and meaningful.

Once the data is prepared, Tableau connects to the database to create a series of visualizations that analyze various aspects of heart disease. These include demographic comparisons, lifestyle impact analysis, and relationships between medical indicators such as BMI, diabetes, and stroke history. Each visualization focuses on a specific analytical objective, and together they provide a comprehensive overview of the dataset.

The dashboards are designed to be interactive, allowing users to apply filters, drill down into specific segments, and explore data from different perspectives. This interactivity enhances user engagement and makes the insights more actionable. For example, doctors can identify high-risk patients, while policymakers can observe trends across populations.

To extend accessibility, the solution also supports web integration using a lightweight framework such as Flask. This enables dashboards and stories to be embedded into a web interface, allowing users to access insights through a browser without needing direct access to Tableau software. The proposed solution is scalable and adaptable, meaning it can be extended with additional datasets or new analytical features in the future. By combining data storage, processing, visualization, and web integration, the system provides a comprehensive platform for heart disease analysis and awareness.